MILITARY SPECIFICATION

CAPACITORS, FIXED, ELECTROLYTIC, DOUBLE LAYER, CARBON (METRIC)

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for double layer, carbon electrode nonsolid electrolytic, fixed capacitors, polarized and nonpolarized, nonhermetically sealed in metal and nonmetal cases. Capacitors covered by this specification may be used to provide a source of reserve energy to prevent erasure of memory-type circuits resulting from primary power interruptions.

1.2 Classification.

1.2.1 Military part numbers. Capacitors specified herein (see 3.1) shall be identified by a military part number which consists of the basic number of the specification followed by a series of coded characters. Each military specification sheet covers a different capacitor style. The coded number shall provide information concerning the capacitor's characteristic, working voltage, capacitance, and capacitance tolerance.

D29501	A	01	B	105	V
Military specification number	Modification (1.2.1.1)	Specification sheet number	Voltage (1.2.1.2)	Capacitance value (1.2.1.3)	Capacitance tolerance (1.2.1.4)

1.2.1.1 Modification. The modification letter is used to indicate the latest characteristic of the part in the specification sheet. This letter shall be changed under the direction of the preparing activity.

1.2.1.2 Voltage. The rated voltage for continuous operation from -40°C to +70°C is identified by a single letter as shown in table I.

TABLE I. Voltage.

Symbol	Rated volts, dc	Surge volts, dc
A- - - - -	1.8	2.2
B- - - - -	5.5	6.3

1.2.1.3 Capacitance value. The nominal capacitance value expressed in microfarads (μF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 8111, Department of the Navy, Washington, DC 20363 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.4 Capacitance tolerance. The capacitance tolerance is identified by a single letter as shown in table II.

TABLE II. Capacitance tolerance.

Symbol	Capacitance tolerance
	Percent
V- - - - -	-20, +80

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Tin Alloy, Lead-Tin Alloy, and Lead Alloy.

MILITARY

MIL-F-14256 - Flux, Soldering, Liquid (Rosin Base).
DOD-C-29501/1 - Capacitors, Fixed, Electrolytic, Double Layer, Carbon, (Nonpolarized), Style DL01 (Metric).
DOD-C-29501/2 - Capacitors, Fixed, Electrolytic, Double Layer, Carbon, (Polarized), Style DL02 (Metric).
DOD-C-29501/3 - Capacitors, Fixed, Electrolytic, Double Layer, Carbon, (Polarized), Style DL03 (Metric).
MIL-C-39028 - Capacitors, Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1276 - Leads for Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 Qualification. Capacitors furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product. The use of paper products is not acceptable.

3.3.1 Soldering flux (when applicable). No acid or acid salts shall be used in the preparation or during soldering the terminals. Soldering flux shall be rosin, rosin and alcohol, or rosin and turpentine, and in accordance with type R or RMA of MIL-F-14256 or QQ-S-571.

3.4 Design and construction. Capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Case. Each capacitor shall be enclosed in a nonhermetically-sealed case, which will prevent leakage or evaporation of the electrolyte and, in addition, will protect the capacitor element against entry of contaminants.

3.4.2 Sleeving. The sleeving shall overlap the ends of the capacitor body.

3.4.3 Solder pin. Solder-pin terminals shall be of a solid conductor of the shape specified (see 3.1).

3.4.4 Wire lead Wire-lead terminals shall conform to MIL-STD-1276 and shall be of a solid conductor of the length and diameter specified (see 3.1).

3.5 Voltage aging. When tested as specified in 4.7.2, capacitors shall withstand the application of voltage and temperature without visible damage.

3.6 DC leakage (DCL). When measured as specified in 4.7.3, the dc leakage shall not exceed the applicable value specified (see 3.1).

3.7 Capacitance. When measured as specified in 4.7.4, the capacitance shall be within applicable tolerance specified (see 3.1).

3.8 Equivalent series resistance (ESR). When measured as specified in 4.7.5, the ESR (in ohms) shall not exceed the value specified (see 3.1).

3.9 Shock (specified pulse). When capacitors are tested as specified in 4.7.6, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, no arcing or other indication of breakdown, nor shall there be any open- or short-circuiting, or evidence of mechanical damage. After testing there shall be no evidence of arcing breakdown or mechanical damage.

3.10 Vibration, high frequency. When capacitors are tested as specified in 4.7.7, there shall be no intermittent contacts of 0.5 ms or greater duration, no arcing or other indication of breakdown, nor shall there be any open- or short-circuiting, or evidence of mechanical damage.

3.11 Vibration. When capacitors are tested as specified in 4.7.8, there shall be no intermittent contacts of 0.5 ms or greater duration, no open- or short-circuiting, nor shall there be any evidence of mechanical damage.

3.12 Thermal shock. When capacitors are tested as specified in 4.7.9, the following requirements shall be met:

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DC leakage	Shall not exceed the initial limit (see 3.1).
Capacitance	Shall change not more than ±10 percent from the initial measured value (see 3.7).
ESR	Shall not exceed the initial limit (see 3.1).
Visual examination	There shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking.

3.13 Resistance to soldering heat. When capacitors are tested as specified in 4.7.10, the following requirements shall be met:

Capacitance	Within applicable tolerance specified (see 3.1).
ESR	Shall not exceed the value specified (see 3.1).
Visual examination	There shall be no evidence of mechanical damage.

3.14 Terminal strength. When capacitors are tested as specified in 4.7.11, there shall be no loosening of the terminals or permanent damage to the terminals.

3.15 Moisture resistance. When tested as specified in 4.7.12, capacitors shall meet the following requirements:

Capacitance	Shall change not more than ±15 percent from the initial measured value (see 3.1 and 3.7).
ESR	Shall not exceed the initial limit (see 3.1).
Visual examination	There shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking wherever applicable.

3.16 Stability at low and high temperatures. When tested as specified in 4.7.13, capacitors shall meet the following requirements:

Step 1 (+25°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall be within tolerance of the nominal value specified (see 3.1).
ESR	Shall not exceed the applicable value specified (see 3.1).

Step 2 (-40°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
ESR	Shall not exceed the applicable value specified (see 3.1).

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Step 3 (+25°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than ±20 percent from the step 1 measured value.
ESR	Shall not exceed the initial limit specified (see 3.1).

Step 4 (+70°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
ESR	Shall not exceed the requirements specified (see 3.1).

Step 5 (+25°C):

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the applicable value specified (see 3.1) from the step 1 measured value.
ESR	Shall not exceed the applicable value specified (see 3.1).

3.17 Surge voltage. When tested as specified in 4.7.14, capacitors shall meet the following requirements:

DC leakage	Shall not exceed the applicable value specified (see 3.1).
Capacitance	Shall change not more than the applicable value specified (see 3.1) from the initial measured value.
ESR	Shall not exceed the initial limit specified (see 3.1).

3.18 Life. When tested as specified in 4.7.15, there shall be no evidence of harmful corrosion, mechanical damage, or obliteration of marking and capacitors shall meet the following requirements:

DC leakage (at +25°C)	Shall not exceed the applicable value specified (see 3.1).
Capacitance (at +25°C)	Shall change not more than the applicable value specified (see 3.1) from the initial measured value.
ESR	Shall not exceed the applicable value specified (see 3.1).

3.19 Solderability. When capacitors are tested as specified in 4.7.16, the criteria for evaluation shall be as specified in method 208 of MIL-STD-202.

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4 GENERAL REQUIREMENTS

3.20 Resistance to solvents. When tested as specified in 4.7.17, the marking shall remain legible and shall not smear, and capacitors shall meet the following requirements:

DC leakage - - - - - - - - - - - -	Shall not exceed the initial limit specified (see 3.1).
Capacitance- - - - - - - - - - - -	Shall change not more than ±5 percent from the initial measured value.
ESR- - - - - - - - - - - - - - - -	Shall not exceed the initial limit specified (see 3.1).

3.21 Marking. Capacitors shall be marked in accordance with method I of MIL-STD-1285. Paper labels shall not be used. Marking shall remain legible after all tests. Capacitor marking shall include the specified information, in the sequence shown in the following example:

D29501A	Military part number
01B105V	
8625A	Date and lot codes (year, week, lot)
12345	Source code (FSCM)
1.0F 5.5 V	Capacitance and rated voltage

Additional marking may appear at the option of the manufacturer provided that it does not interfere with the required marking.

3.22 Workmanship. Capacitors shall be processed in such a manner that when examined under 10X magnification, they shall be uniform in quality and shall be free from pits, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.22.1 Soldering. All excess flux or solder shall be removed. Electrical connections shall be electrically continuous after soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment, and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Retention of qualification (see 4.5).
c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

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4.3.2 AC measurements. AC measurements shall be made at the frequency specified. The magnitude of the ac voltage shall be equal to or less than 1.0 volt root mean square (rms).

4.3.3 Reference measurements. When requirements are based on comparative measurements made before and after conditioning, the reference measurement shall be considered the last measurement made at +25°C ±5°C prior to conditioning. Unless such measurements have been made within 30 days prior to the beginning of conditioning, they shall be repeated.

4.3.4 Power supply. The power supply used for life testing shall have a regulation of ±2 percent or less of the rated voltage. The power source employed for dc leakage current measurements shall be stablilized to at least ±100 parts per million. During measurements there must be no voltage fluctuations of sufficient amplitude to produce a variation in the current measurement as read with any dc leakage current tester used to test capacitors.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number and style combinations of capacitors to be subjected to qualification inspection shall be as specified in the appendix to this specification.

4.4.2 Inspection routine. The sample units shall be subjected to the inspections specified in table III, in the order shown.

4.4.3 Failures. Failures in excess of those allowed in table III shall be cause for refusal to grant qualification approval.

4.5 Retention of qualification. To retain qualification, the contractor shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (groups A and B), indicating as a minimum, the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
b. A summary of the results of tests performed for periodic inspection (group C), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 6-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 6-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of the specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during three consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each style to testing in accordance with the qualification inspection requirements.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspections. Except as specified in 4.6.1.4.3, delivery of products which have passed groups A and B inspections shall not be delayed pending the results of group C inspection.

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4.6.1.1 Inspection lot. An inspection lot shall consist of all capacitors from the same production line or lines, of the same style (see 3.1), produced under essentially the same conditions with the same basic materials, and offered for inspection during a single month. The capacitance values and voltages produced shall be represented in the lot in approximately the same ratio of production.

TABLE III. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group I				
Voltage aging	3.5	4.7.2	154	0
DC leakage	3.6	4.7.3		
Capacitance	3.7	4.7.4		
ESR	3.8	4.7.5		
Visual and mechanical inspection	3.1, 3.3, 3.3.1, 3.4, 3.4.1, 3.4.2, 3.21 and 3.22	4.7.1		
Group II				
Shock (specified pulse)	3.9	4.7.6	12	1
Vibration, high frequency 1/	3.10	4.7.7		
Vibration 2/	3.11	4.7.8		
Thermal shock	3.12	4.7.9		
Group III				
Resistance to soldering heat	3.13	4.7.10	18	
Terminal strength	3.14	4.7.11		
Moisture resistance	3.15	4.7.12		
Group IV				
Stability at low and high temperatures	3.16	4.7.13	12	
Surge voltage	3.17	4.7.14		
Group V				
Life (1,000 hours at +70°C)	3.18	4.7.15	102	1
Group VI				
Solderability	3.19	4.7.16	10	0
Resistance to solvents 3/	3.20	4.7.17		

1/ For style DL01.
2/ For styles DL02 and DL03.
3/ Style DL01 will check for marking only, styles DL02 and DL03 will check marking and electrical tests.

4.6.1.2 Prescreen inspection. All capacitors submitted for group A inspection shall be 100 percent prescreened for the inspections specified in table IV, in the order shown.

TABLE IV. Prescreen inspection.

Inspection	Requirement paragraph	Test method paragraph
Voltage aging	3.5	4.7.2
DC leakage	3.6	4.7.3
Capacitance	3.7	4.7.4
ESR	3.8	4.7.5

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4.6.1.3 Group A inspection. Group A inspection shall consist of the inspections specified in table V, in the order shown.

TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (%) Major	AQL (%) Minor
Subgroup 1				
Capacitance	3.7	4.7.4	0.65	
ESR	3.8	4.7.5		
Solderability 1/	3.18	4.7.15		
Subgroup 2				
Visual and mechanical inspection	3.1,3.3,3.3.1,3.4, 3.4.1,3.4.2,3.21 and 3.22	4.7.1	1.0	4.0

1/ Eight pieces to be inspected, one failure allowed.

4.6.1.3.1 Sampling plan. The sampling plan for subgroups 1 and 2 shall be in accordance with MIL-STD-105 for special inspection level II. The acceptable quality levels (AQL) shall be as specified in table V. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.3.2 Rejected lots. Lots rejected by the group A inspection shall be segregated from new lots and those lots that have passed inspection. Lots rejected because of failures in subgroup 2 may be offered for acceptance only if the contractor inspects all units in the lot for those qualifying characteristics found defective in the sample and after removing all defective units found, reinspects the lot using the tightened inspection procedures of MIL-STD-105. Resubmitted lots shall be kept separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.4 Group B inspection. Group B inspection shall consist of the inspections specified in table VI in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE VI. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup 1		
Stability at low and high temperatures	3.16	4.7.13
Subgroup 2		
Surge voltage	3.17	4.7.14

4.6.1.4.1 Sampling plan. The sampling plan for subgroup 1 shall be in accordance with MIL-STD-105 for special inspection level S-4. The AQL shall be 1.5 percent defective. For subgroup 2, 12 sample units taken from subgroup 1 shall be subjected to the test. There shall be no more than one failure.

4.6.1.4.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.4.3 Disposition of sample units. Sample units which have been subjected to subgroup 2, group B inspection, shall not be delivered on the contract.

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4.6.1.5 Group C inspection. Group C inspection shall consist of the inspections specified in table VII, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspections.

4.6.1.5.1 Sampling plan. Sixty-five sample units of each style shall be selected from the first lot produced, and thereafter from each production of 500,000 units or once every 3 months, whichever occurs first. The 65 sample units shall be subdivided as specified for each subgroup in table VII.

4.6.1.5.2 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Units to be inspected	No. of defectives allowed
Subgroup 1				
Shock (specified pulse)	3.9	4.7.6	12	
Vibration, high frequency 1/	3.10	4.7.7		
Vibration 2/	3.11	4.7.8		
Thermal shock	3.12	4.7.9		
Subgroup 2				
Resistance to soldering heat	3.13	4.7.10	18	1
Terminal strength	3.14	4.7.11		
Moisture resistance	3.15	4.7.12		
Subgroup 3				
Resistance to solvents 3/	3.20	4.7.17		
Subgroup 4				
Life (1,000 hours at +70°C)	3.18	4.7.15	25	1

1/ For style DL01.
2/ For styles DL02 and DL03.
3/ Style DL01 will check for marking only, styles DL02 and DL03 will check marking and electrical tests.

4.6.1.5.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall immediately notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

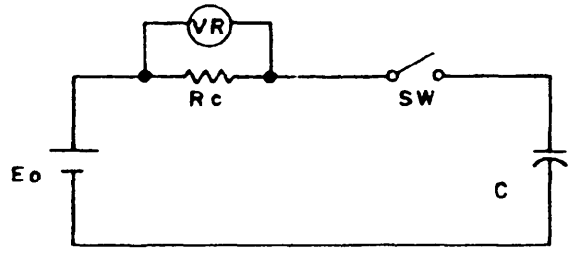
4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. Capacitors shall be inspected to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.4.1, 3.4.2, 3.21 and 3.22).

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4.7.2 Voltage aging (see 3.5). Capacitors shall be subjected to the dc rated voltage, minimum, for 2 hours minimum, at a temperature of +70°C ±5°C. The voltage aging circuit shall have a series resistance of 3.0 ohms maximum. Power source shall not be less than 5 amperes. Capacitors shall then be stabilized at room temperature and examined for evidence of mechanical damage.

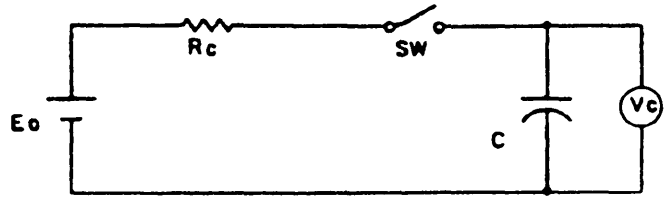
4.7.3 DC leakage (see 3.6). DC leakage shall be measured using the following method:



$$DCL = \frac{VR}{Rc} \text{ (A)}$$

NOTES:
1. Rc (see 3.1)
2. Eo = 5.0 ±0.1 V.
3. VR = voltage drop by resistance Rc 30 minutes for 1 farad and less, 60 minutes for greater than 1 farad after Eo applied.
4. Initial V_c shall be less than or equal to .05 volt.

4.7.4 Capacitance (see 3.7). Capacitance shall be calculated using the following method:



$$C = \frac{\tau}{Rc} \text{ (F)}$$

Where:
Eo = 5.0 ±0.1 V.

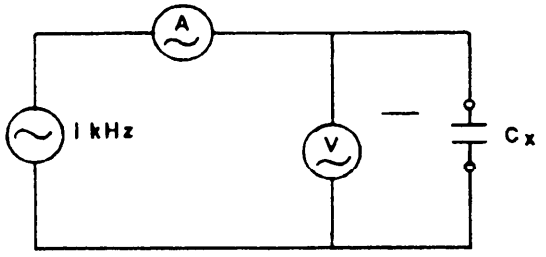
R_c =		
	1 kΩ	(0.022 F, 0.047 F)
	510Ω	(0.1 F)
	200Ω	(0.22 F)
	100Ω	(0.47 F and above)

τ = Time constant, that is, time period from 0 to 0.632 Eo volt.

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4.7.5 Equivalent series resistance (ESR) (see 3.8). The ESR at the applicable temperature shall be measured directly or determined from measurements obtained from a bridge. The following details shall apply:

a. Test temperature and tolerance - Capacitors shall be at +25°C ±5°C.
b. Test frequency - 1,000 ±100 Hz.
c. Limit of accuracy - Measurement accuracy shall be within ±5.0 percent of the reading.
d. Magnitude of voltage - The magnitude of the ac voltage shall be limited to 0.5 volt rms maximum.



Where:
Cx: Capacitor
Ⓐ: Ampere meter (ac)
Ⓥ: Volt meter (ac)
(Impedance shall be sufficiently higher that that of Cx.)

4.7.6 Shock (specified pulse) (see 3.9). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting means - Capacitors shall be rigidly mounted on a mounting fixture by the body. (Potting compounds may be used to secure the capacitor body as long as the compound does not support the leads.) Leads shall be secured to rigidly supported terminals, spaced so that the length of each lead from the capacitor is approximately 10 mm when measured from the edge of the supporting terminal. Leads shall be within 30 degrees of being parallel. When securing leads, care shall be taken to avoid pinching the leads.
b. Test condition letter - G (50 g peak).
c. Measurements and electrical leading during shock - During the test, observations shall be made to determine intermittent contact, arcing or open- or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms. The dc rated voltage (see 3.1) shall be applied to the capacitors during the test.
d. Examinations after test - Capacitors shall be visually examined for evidence of arcing, breakdown, and mechanical damage.

4.7.7 Vibration, high frequency (see 3.10). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens - Capacitors shall be mounted on a fixture by the body. Leads shall be supported by rigidly supported terminals.
b. Electrical load conditions - During the test, the specified dc rated voltage (see 3.1) shall be applied to the capacitors.
c. Test condition letter - D (20 g).
d. Duration and direction of motion - 4 hours in each of 2 mutually perpendicular planes (total of 8 hours), one parallel to the principal axis.
e. Measurements during vibration - During the last cycle, an electrical measurement shall be made to determine intermittent operation or open- or short-circuiting. Observations shall also be made to determine intermittent contact, arcing or open- or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms, or greater.
f. Measurements after vibration - Not applicable.
g. Examination after test - Capacitors shall be visually examined for evidence of mechanical damage.

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4.7.8 Vibration (see 3.11). Capacitors shall be tested in accordance with method 201 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting of specimens - Capacitors shall be mounted on a fixture by the body. Leads shall be supported by rigidly supported terminals.
b. Electrical load conditions - During the test, the specified dc rated voltage (see 3.1) shall be applied to the capacitors.

4.7.9 Thermal shock (see 3.12). Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter - A.
b. Measurements after thermal shock - DC leakage, capacitance, and ESR shall be measured as specified in 4.7.3, 4.7.4, and 4.7.5, respectively.
c. Examination after test - Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking (wherever applicable).

4.7.10 Resistance to soldering heat (see 3.13). Capacitors shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter - B. Procedure 1, leads shall be immersed to the depth of 1.5 mm from the end seal or capacitor body.
b. Cooling time prior to final examinations and measurements after test - 1.0 hour minimum.
c. Examination after test - Capacitance and ESR shall be measured as specified in 4.7.4 and 4.7.5, respectively. Capacitors shall be visually examined for evidence of mechanical damage.

4.7.11 Terminal strength (see 3.14). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply, unless otherwise specified (see 3.1):

a. Test condition letter - A; the body of the capacitor shall be secured; 20 newtons applied force.
b. Test condition letter (wire leads only) - C; 20 newtons applied force; 3 bends.
c. Measurements after test - After the test, capacitors shall be visually examined for loosening of the terminals and permanent damage to the terminals.

4.7.12 Moisture resistance (see 3.15). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting - Special mounting not required.
b. Initial measurements - Capacitance as specified in 4.7.4.
c. Number of cycles - Ten continuous cycles except steps 7a and 7b are not required.
d. Loading voltage - Not applicable.
e. Final measurements - After removal from chamber, capacitors shall be dried for 1 hour at room temperature and within 2 to 6 hours, capacitance and ESR shall be measured as specified in 4.7.4 and 4.7.5, respectively.
f. Examination after test - Capacitors shall be visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking.

4.7.13 Stability at low and high temperatures (see 3.16). DC leakage, capacitance, and ESR shall be measured as specified in 4.7.3, 4.7.4, and 4.7.5, respectively, at each of the temperatures specified in table VIII except that dc leakage measurements at -40°C (step 2) are not required. The capacitors shall be brought to thermal stability at each test temperature. Thermal stability has been reached when no further change in capacitance is observed between two successive measurements taken at intervals of 15 +2, -0 minutes. Capacitors may be dried at +70°C for 30 +4, -0 minutes prior to start of tests. The decision to reject the parts shall be made on those parts that have been dried.

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TABLE VIII. Temperature for stability test.

Step	Test temperature (°C)
1	+25 ±2
2	-40 +0, -3
3	+25 ±2
4	+70 +4, -0
5	+25 ±2

4.7.14 Surge voltage (see 3.17). Capacitors shall be subjected to 1,000 cycles of the applicable surge voltage specified in table I. The ambient temperature during cycling (see 3.1), shall be +70°C ±5°C. Each cycle shall consist of a 30 +2, -0 second surge voltage application followed by a 30 +2, -0 second discharge period. Voltage application shall be made through a resistor of 33 ohms. The tolerance of the resistor shall be ±5 percent. Each surge voltage cycle shall be performed in such a manner that the capacitor is shorted terminal to terminal through a copper bar, or an equivalent low resistance at the end of the 30 +2, -0 second application. An alternate method of shorting the capacitors is discharge through the same resistance that is utilized for charging. After the final cycle, the capacitors shall be stabilized at the inspection conditions specified in 4.3, and the dc leakage, capacitance, and ESR shall be measured as specified in 4.7.3, 4.7.4, and 4.7.5, respectively.

4.7.15 Life (see 3.18). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Test temperature and tolerance - Capacitors subjected to the test of group V of table III shall be tested at +70°C +4, -0.
b. Operating conditions - DC rated voltage (see 3.1), shall be applied gradually (not to exceed 5 minutes either by a slow buildup of the voltage or through a resistor which shall be shorted out within 5 minutes). Voltage shall be applied continuously, except for measurement periods. The impedance of the voltage source, as seen from the terminals of each capacitor, shall not exceed 3 ohms. Storage batteries or an electronic power supply capable of supplying at least 1 ampere when a capacitor is shorted shall be used.
c. Test condition letter - D (1,000 hours).
d. Measurements during exposure - DC leakage shall be measured as specified in 4.7.3 at 0; 240 +48, -0 hours; 1,000 +48, -0 hours. The manufacturer may option to measure dc leakage at +70°C; however, if the capacitors fail to meet the dc leakage requirement in 3.17, the measurement shall be made at +25°C.
e. Measurements after exposure - Capacitors shall be returned to the inspection conditions specified in 4.3 and visually examined for evidence of harmful corrosion, mechanical damage, and obliteration of marking (if applicable); dc leakage, capacitance and ESR shall then be measured as specified in 4.7.3, 4.7.4, and 4.7.5, respectively.

4.7.16 Solderability (see 3.19). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. The number of terminations of each capacitor to be tested - Two.
b. Depth of immersion in flux and solder - All terminals shall be immersed to within 3 mm of the capacitor body.

4.7.17 Resistance to solvents (see 3.20). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following exception shall apply:

a. Measurements and examination after test - DC leakage, capacitance, and ESR shall be measured as specified in 4.7.3, 4.7.4, and 4.7.5, respectively.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-39028.

6. NOTES

6.1 Intended use. Capacitors covered by this specification are intended mainly for use as a standby power source for low-power memories during power outages lasting up to several days. Since these capacitors have relatively high ESR (0.5 to 50 ohms), they are not recommended for filtering applications. These capacitors should not be used in critical applications or where extreme temperatures will be encountered.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of the applicable specification sheet, and the complete type designation (see 3.1).

6.3 Qualification. 1/ With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the qualified products list is Space and Naval Warfare Systems Command; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444.

6.4 Subject term (key word) listing.

Capacitor
Carbon
Double layer
Electrolytic
Memory-type circuits
Nonhermetically sealed

1/ SD-6, "Provisions Governing Qualification" is issued for the information of applicants requesting qualification of products. Copies of this publication may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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APPENDIX

PROCEDURES FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the qualification-submittal plans for capacitors to be subjected to the qualification inspection specified in this specification. These plans specify the sample size, sampling criteria, and limits of coverage for the type of qualification sought.

10.2 Qualification category. Qualification shall be categorized as follows:

Category A - Qualification shall be in accordance with the requirements of this specification.

20. SUBMISSION

20.1 Sample.

20.1.1 Single-type submission. A sample consisting of 154 units, as applicable (see tables III and IX), sample units of each style in the highest capacitance value in the voltage rating for which qualification is sought shall be submitted. The capacitance tolerance shall be the same for all submitted samples.

20.2 Test data. When inspections are to be performed at a Government laboratory, prior to submission, all sample units shall be subjected to all of the nondestructive tests in table I. Each submission shall be accompanied by the test data obtained from these examinations and tests. The performance of the destructive examinations and tests by the manufacturer on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 Certification of material. When submitting samples for qualification, the manufacturer shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

20.4 Description of items. The manufacturer shall submit a detailed description of the capacitors being submitted for inspection including information on the type of welds or solder buttons, the type of electrodes, the type of seals (inner or outer, as applicable), the case and lead materials, the case insulating material (when applicable), and the case finish.

TABLE IX. Samples for complete qualification submissions.

Types to be submitted	Quantity	When qualification is sought for
D29501A01B223V	154	Style DL01. All capacitance values.
D29501A02B104V	154	Style DL02. All capacitance values.
D29501A03A334V	154	Style DL03. All capacitance values.

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Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AV
Navy - SH
Air Force - 17, 85, 99
DLA - ES

User activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5910-1449)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-C-29501	Capacitors, Fixed, Electrolytic, Double Layer, Carbon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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